



reduces infrastructure expenses but also alleviates the strain on our precious energy resources. Its benefits extend further, streamlining infrastructure demands and reducing upfront investments and maintenance costs, enabling rapid deployment, especially in regions lacking water infrastructure. Envision a future where long-distance water transportation becomes obsolete, revolutionising access to water in remote and arid areas, offering substantial cost savings and environmental benefits. Water-from-air technology is versatile, serving individual households, large community projects, and disaster relief efforts alike. Join us in exploring this remarkable journey, where innovation meets necessity, and the possibilities are as vast as the atmosphere itself.

This visionary approach not only champions eco-friendliness but also provides a blueprint for long-term cost-effectiveness, protecting both our ecology and economy. Advanced filtration and purification technologies ensure that the water produced meets and often exceeds the strictest quality standards. This eliminates the need for additional treatment processes, saving time and resources. While there may be initial investment costs, the long-term gains and sustainability of water-from-air technology make it an appealing solution for addressing global water challenges. It stands as a beacon of hope for a world where clean, accessible water is not just a distant dream but a reality for all. Imagine a future where water innovation harmonises with the environment, and the possibilities are boundless.

Researchers at Southern Federal University (SFedU) in Russia have developed an autonomous device that extracts water from the air using a metal-organic framework polymer sorbent. This sorbent absorbs water vapour at night and releases

Case studies of Vayujal

1. Institutional consumer. At Gram Vikas in Mohuda, Odisha, Vayujal's water generation units were tested under both grid-powered and solar-powered conditions, each producing 400 litres per day (LPD). The grid-powered unit operated with an average humidity of 73.61% and an average temperature of 28.40°C, generating a total of 7035 litres of water with an efficiency of 0.273kWh per litre. The solar-powered unit, on the other hand, functioned under 70.82% humidity and a temperature of 32.24°C, producing 6915 litres with an efficiency of 0.284kWh per litre.

2. Hospitality segment. Since April 2021, the ITC-Chola hotel in Guindy, Chennai, has been using a Vayujal unit powered by in-house wind energy and supported by an in-house bottling plant. This unit operates at an average relative humidity of 70% and an average temperature of 30°C. With an electricity cost of ₹0.6 per kWh, it has generated 88,200 litres of water. The unit's power efficiency is notable at 0.27kWh per litre, underscoring its effective use of wind energy for sustainable water production.

3. IT park segment. At Engineers India Limited in Gurugram, a Vayujal unit has been operational since October 2022. It is powered by an on-grid solar power plant installed on the rooftop. The unit operates with an average relative humidity of 45% and an average temperature of 35°C. Despite these challenging conditions, it has generated 68,200 litres of water, with an electricity cost of ₹0.6 per kWh. Its power efficiency stands at 0.32kWh per litre, demonstrating the unit's capability to efficiently convert solar energy into water.

4. Tourism industry. Since April 2023, Vayujal units have been deployed at three locations under the Tamil Nadu Tourism Development Corporation (TTDC). These units are powered by a mix of grid and solar power sources. They maintain an average relative humidity of 60% and operate in environments averaging 30°C in temperature. Despite a higher electricity cost of ₹8 per kWh, these units have collectively generated 48,700 litres of water. They exhibit a power efficiency of 0.26kWh per litre, reflecting their efficiency in water production using diverse power sources.

it during the day when heated by sunlight. Currently undergoing field tests in southern Russia, the device aims to address drinking water shortages in regions lacking infrastructure, such as parts of Africa and Southeast Asia. Ilya Pankin, head of the laboratory, highlighted its ability to function even at humidity levels below 25%, with a cost of 9-10 rubles per litre (\$0.12-0.14). Inspired by African beetles, the device operates on solar energy, providing a sustainable and energy-efficient solution to water scarcity.

Uravu Labs emphasises harnessing solar energy in their operations, using systems like residential solar water heaters. This approach not only reduces their reliance on electricity but also significantly lowers the cost of producing water from the air. By integrating solar and waste heat sources, the cost drops from around 4-5 rupees per litre to just 1.5-2 rupees per litre—a substantial

difference. Uravu Labs caters to diverse needs, establishing micro-factories that produce up to 10,000 litres of water daily, and offering packaging options ranging from 500 millilitre bottles to bulk supply.

The cost of water production varies based on energy sources and local electricity rates. Vayujal's technology offers a bespoke cost-efficiency formula for each location, measured in energy-per-litre. In corporate offices benefiting from discounted electricity rates, the cost per litre can drop to two to three rupees. In regions with electricity subsidies, such as government schools or hospitals, the cost hovers around one rupee per litre.

Majik Water's innovative energy pricing model further highlights the affordability of water production. According to Nyoro, their most significant operational unit, capable of producing 1000 litres daily, consumes just 16kWh of energy. At an assumed electricity rate of ₹8.50